ROQ: a new measure for more accurately filtering mapped reads

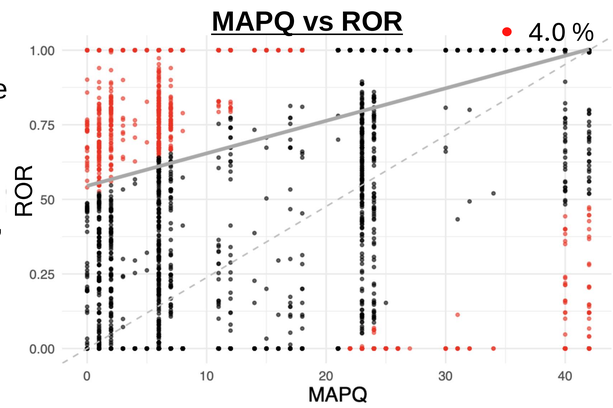
Nayoung Park[1], Minji Gu[1], and Jaebum Kim[1]
[1]*Department of Biomedical Science and Engineering, Konkuk University*

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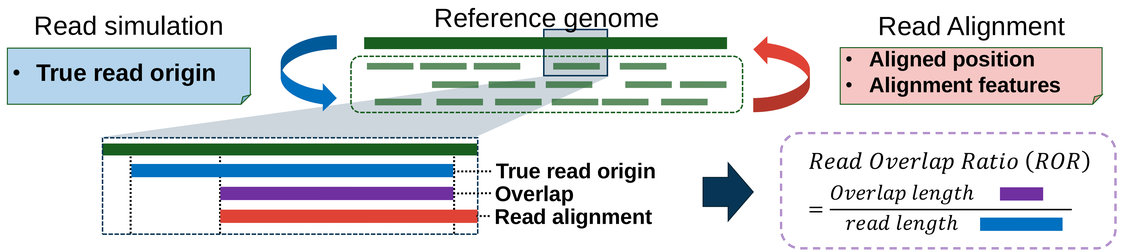
Problem

Next-generation sequencing (NGS) has become essential in genomic research, but assessing the accuracy of read alignment remains challenging. The widely used Mapping Quality (MAPQ) score often fails to reflect the true origin of sequencing reads, especially in complex genomic regions. This limitation can lead to errors in downstream analyses, such as variant detection and gene expression quantification. A more robust and precise metric is needed to accurately evaluate alignment quality, improving the reliability of genomic insights.



Approach

ROQ simulation data generation workflow



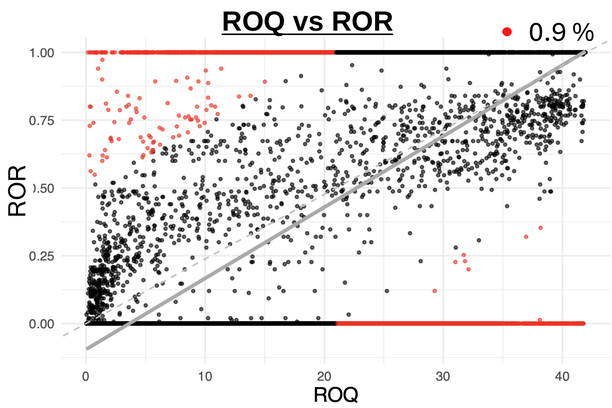
Feature information

Feature	Description
MAPQ	Probability that a read is correctly aligned to the reference genome.
EDIT_DIST	Total number of operations (insertions, deletions, mismatches) needed to alignment.
SECONDARY_ALIGN_SCORE	Alignment score for the best secondary (non-primary) alignment.
MISMATCHES	Number of base mismatches between the read and the reference sequence.
MATE_ALIGNMENT_SCORE	Alignment score for the mate of a paired-end read.
ALIGN_SCORE	Score based on how well the read matches the reference.
ALIGN_SCORE_DIFF	Difference between the primary and secondary alignment scores.
READ_GC_CONT	Proportion of guanine (G) and cytosine (C) bases in the read.
INSERT_SIZE	Inferred insert size (or fragment length) between paired-end reads.
GAP_OPENS	Number of gap openings (insertions or deletions) introduced in the alignment.
GAP_EXT	Total number of gap extensions after the initial gap opening.
AVG_QUALITY_BASE_SCORE	Average base quality score across the read.
N_LOW_QUALITY_BASE	Number of bases in the read with a quality score below a certain threshold.

ROQ model

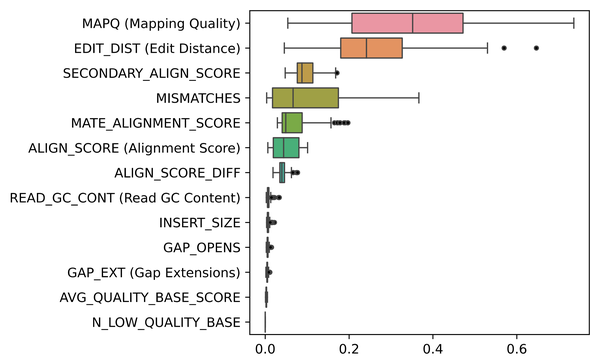


Results





Feature Importance



Various machine learning models

Machine learning algorithms	RMSE	MAE	R^2	Corr. coef.
Linear Regression	0.1214	0.0396	0.4813	0.6938
Ridge Regression	0.1214	0.0396	0.4813	0.6938
Lasso Regression	0.166	0.0569	0.0308	0.5751
Decision Tree Regressor	0.1181	0.0153	0.5094	0.7497
Random Forest Regressor	0.095	0.0154	0.6824	0.8277
Gradient Boosting Regressor	**0.0892**	0.0167	0.7197	0.8484
XGBoost Regressor	0.0876	**0.0158**	**0.7298**	**0.8543**

Discussion

The ROQ model provides a more reliable metric for read alignment quality than MAPQ, demonstrating robust performance across simulated datasets, diverse genomic regions, and different species. This improvement in alignment accuracy can reduce errors in downstream analyses, such as variant detection and gene expression quantification, enhancing the reliability of genomic insights.

While these initial results are promising, further validation with real-world data is needed to confirm the model's robustness under varied sequencing technologies and experimental conditions. Future work will focus on refining the ROQ model, integrating additional alignment features, and expanding its applicability across broader genomic research applications.

Acknowledgements

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References

Cline E, Wisittipanit N, Boongoen T, Chukeatirote E, Struss D, Eungwanichayapant A. Recalibration of mapping quality scores in Illumina short-read alignments improves SNP detection results in low-coverage sequencing data. PeerJ. 2020 Dec 7;8:e10501. doi: 10.7717/peerj.10501. PMID: 33354434; PMCID: PMC7727374.